

Kubernetes

Kubernetes: concept, Quality Engineering use cases, first steps and official reference.

What is it?

Kubernetes is part of the modern software engineering and quality ecosystem. This guide explains its role, where it fits in the delivery flow, and how QA professionals can use it with the right context.

Why should QA know it?

Understanding Kubernetes helps QA collaborate with development, DevOps, data and product teams. The goal is not to memorize commands, but to understand how the tool affects testability, automation, evidence, diagnostics and delivery quality.

Getting started

Start with the official documentation, verify prerequisites and install only from the official source. In corporate environments, confirm internal policies, supported versions, permissions and environment variables before configuring Kubernetes.

Practical QA use

Use Kubernetes in a small reproducible exercise. Record the goal, configuration, expected result, evidence and issues. Then connect the learning to a real testing, automation, integration, data or continuous-delivery scenario.

Good practices

Prefer official documentation, never expose credentials, version configuration when appropriate, keep dependencies current and document important decisions. A tool without context does not replace a quality strategy.

When to use it

Use this knowledge when product risk, architecture or the delivery stage requires this type of validation. Confirm the goal, environment, required data and expected evidence before starting.

QA tip

Always connect the practice to product risk and the evidence needed for quality decisions.

Official resources

Site oficial: <https://kubernetes.io/docs/setup/>